

Final Prototype Report

Ethan Kelly - 100789657, Clover Espiritu-santo - 100829486, Jevin Chen - 100819841

Faculty of Science, Ontario Tech University

CSCI 4620U Human-Computer Interaction

Dr. Ali Neshati

December 1st, 2025

Overview

We chose to break our project into two separate prototypes: one for the desktop application using Figma to implement the UI, and one for the web extension using HTML/JS to implement the main functionality of the project.

Desktop

This prototype uses Figma to create a web app that can edit any academic documents. It demonstrates a general look and feel as well as some functionality that may be added to the final application.

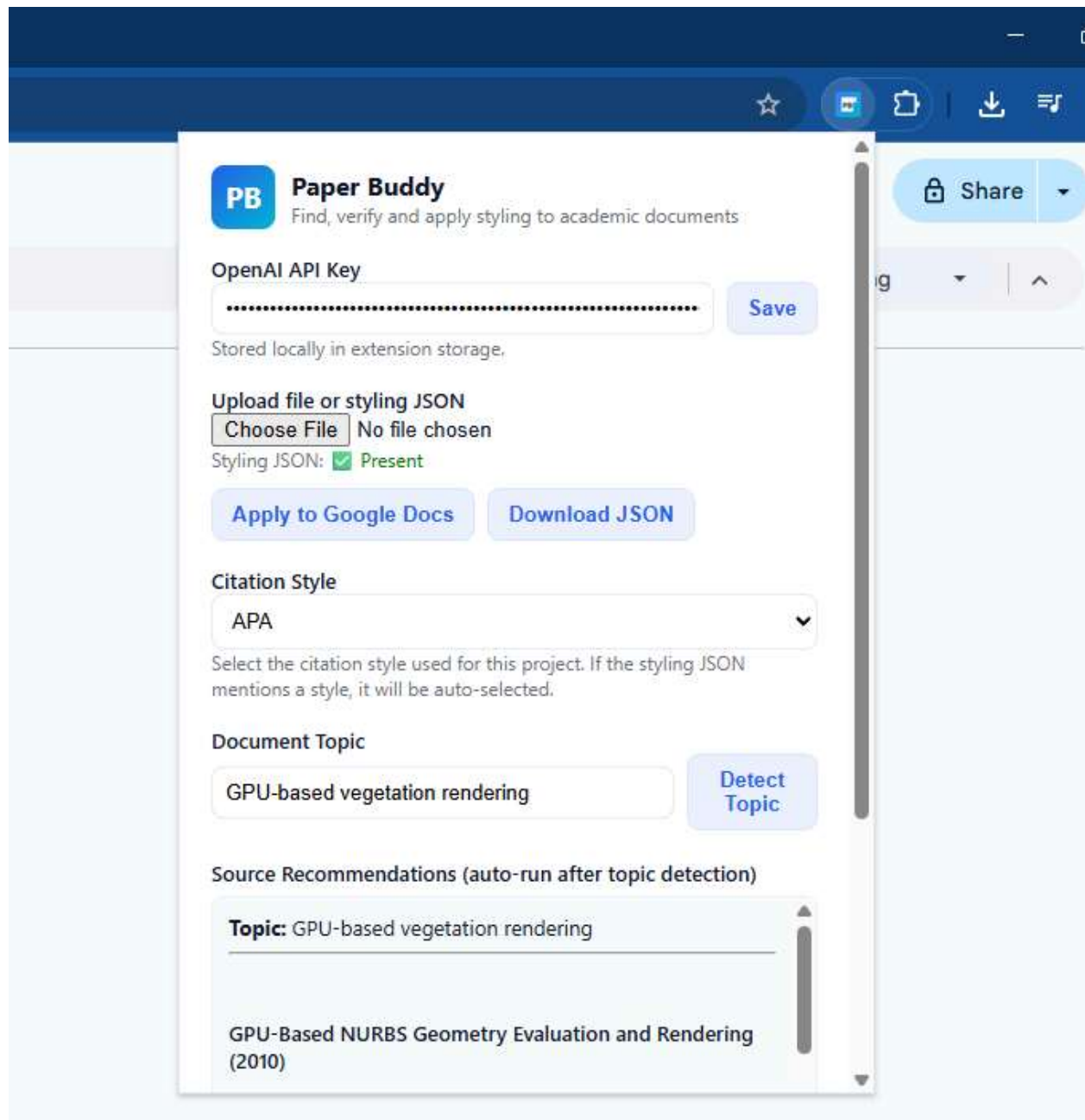
The prototype demonstrates the look and feel of adding a suggested citation to the paper as well as an easy way of adding any formatting changes that may arise streamlining the writing process for academics and students.

Web

This prototype uses HTML and JS to create a working Chrome extension following the intended functionality of the project without much consideration for the UI due to the constraints on our group. To quote Top Gear: due to the unique way our project is funded the user must provide their own OpenAI API key to activate the AI functionality of the project. This is stored locally in session storage.

The user can upload a file to the extension, which is then analyzed with AI to find styling guidelines and compressed into a custom json object. The user can download this object to reupload later, or apply the settings to an open Google Docs tab. Applying the styling json to an open tab is currently incomplete functionality. This json also auto-selects the correct citation style if one is provided in the instructions.

Users can also use the Paper Buddy web prototype to analyze their writing. Clicking Detect Topic uses AI to analyze the content of the page and output a concise document topic. As a prototype, the first run of this functionality doesn't always work, but successive attempts do work in testing. Once a topic is found, Paper Buddy automatically finds three real academic articles to recommend to the user. I want to be very clear that this does actually work and recommends real relevant articles to the topic of the paper. It also includes automatic citations in the correct style if a style is chosen.



Video Link

<https://drive.google.com/drive/folders/1kfigKBz5KrLPCBsnHnz6EHY2L27kwSPY?usp=sharing>

Appendix

Ethan Kelly	Built the entire chrome extension prototype.
Jevin Chen	Built the entire desktop prototype.
Clover Espiritu-Santo	Designed overall UI and functionality requirements